Indian Institute of Technology Madras Zanzibar Campus
Liberal Democracy

Index to measure Health of Judiciary

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Find the code work used in our GitHub Repo!

1 Introduction

The judiciary is one of the most important pillars of democracy. It ensures laws are upheld, governments accountable, and protects rights.

A strong judiciary is essential for political stability and the rule of law, but its effectiveness can vary significantly between countries. Factors such as political interference, corruption, and access to justice all influence how well a judicial system functions.

This study measures the **Health of the Judiciary** using a combination of multiple indicators sourced from the **V-Dem dataset**. The goal is to capture different aspects of judicial performance.

To create a meaningful assessment, we used two approaches:

1. A **Weighted Judiciary Health Index**, where key indicators were normalized and assigned weights based on their significance.
2. A **PCA-Based Judiciary Health Index**, which identifies the most influential factors through statistical decomposition.

We applied these indices to **African countries in 2023** and also analyzed trends over time for **India, Tanzania, Zanzibar, Japan, and Germany** from 2001 to 2023. The findings provide insights into where judicial systems are strong, where they are weak, and how different methods compare in evaluating judiciary health.

2 Methodology

To evaluate the **Health of the Judiciary** across various countries, we developed a **composite index** using two different methodologies: a **weighted index** and **Principal Component Analysis (PCA)**.

2.1 Selection of Indicators

The Judiciary Health Index was constructed using **14 key indicators** from the V-Dem dataset, covering judicial independence, integrity, accountability, transparency, and political interference. We, earlier, considered an additionaly indicator, but this *(Judicial Review)* was removed due to it being just a simple yes or no indicator.

2.1.1 Judicial Independence (0.24)

Indicators: High Court Independence (v2juhcind), Lower Court Independence (v2juncind)

Judicial independence is fundamental to free and democratic decision-making. These indicators measures the autonomy of courts from external influence.

2.1.2 Judicial Accountability and Corruption (0.16)

Indicators: Judicial Corruption Decisions (v2jucorrdc), Judicial Accountability (v2juaccnt)

A judiciary must operate with integrity. These indicators check on the corruption levels, ensuring judicial responsibility.

2.1.3 Legal Transparency and Compliance (0.16)

Indicators: Transparent Laws (v2cltrnslw), Compliance with High Court (v2juhccomp), Compliance with Judiciary (v2jucomp)

Transparency and compliance are essential for legal consistency. These indicators reflect whether judicial rulings are applied aptly.

2.1.4 Judicial Reforms and Political Interference (0.22)

Indicators: Judicial Reform (v2jureform), Judicial Purges (v2jupurge), Government Attacks on Judiciary (v2jupoatck), Court Packing (v2jupack)

Political interference undermines judicial stability. These indicators capture attempts to manipulate the judiciary through attacks or threats, bribes, etc.

2.1.5 Access to Justice and Rule of Law (0.22)

Indicators: Access to Justice for Men (v2clacjstm), Access to Justice for Women (v2clacjstw), Impartial Public Administration (v2clrspct)

A judiciary must be accessible and fair. These indicators ensure that legal systems are open to all citizens, regardless of gender or status.

2.1.6 Weight Justification

Weights were assigned based on importance:
- **Judicial Independence (0.24)**: Most critical for fair rulings.
- **Judicial Accountability and Corruption (0.16)**: Ensures legitimacy.
- **Legal Transparency and Compliance (0.16)**: Reflects rule enforcement.
- **Judicial Reforms and Political Interference (0.22)**: Addresses threats to judicial autonomy.
- **Access to Justice and Rule of Law (0.22)**: Ensures fair access to courts.

These indicators provide a **balanced and comprehensive measure of judicial health**.

2.2 Normalization of Indicators

Each selected indicator follows an **ordinal scale** with different numerical ranges. To ensure comparability, we normalized all indicators to a **0 to 1 scale** by dividing by their respective **maximum ordinal values**:

$$X_{\text{normalized}} = \frac{X}{\text{MaxOrdinal} + 1} \tag{1}$$

Where:

- X is the original indicator value
- MaxOrdinal is the maximum ordinal level of the indicator

2.3 Weighted Index Construction

After normalization, a **weighted sum approach** was used to compute the Judiciary Health Index. Each indicator was assigned a specific weight based on its relevance:

$$JHI_{\text{weighted}} = \sum_{i=1}^{n} w_i X_i \tag{2}$$

Where:

- JHI_{weighted} is the weighted Judiciary Health Index

- X_i represents each normalized indicator
- w_i is the weight assigned to each indicator
- $\sum w_i = 1$ (weights sum to 1)

The assigned weights were carefully chosen based on expert evaluation of the relative importance of each factor in assessing judicial health.

2.4 Principal Component Analysis (PCA)

In addition to the weighted approach, we applied **Principal Component Analysis (PCA)** to create an alternative Judiciary Health Index. PCA transforms the set of indicators into a smaller number of **uncorrelated principal components**, capturing the most significant variance in the data.

The transformation follows:

$$Z = WX \tag{3}$$

Where:

- X is the matrix of standardized indicators
- W is the matrix of eigenvectors (principal components)
- Z is the transformed feature matrix

Only the first principal component (PC1) was retained to define the Judiciary Health Index, as it captures the maximum variance in the dataset. The resulting scores were **normalized to the range [0,1]** for comparability.

2.5 Comparison of Indices

To evaluate the effectiveness of both methods, we conducted a **comparative analysis** between the **weighted index** and the **PCA-based index**. This involved Computing the **Pearson correlation coefficient** to measure their alignment.

We saw A strong correlation ($\mathbf{r = 0.8837}$) that indicated that both methods produced similar rankings of judiciary health across countries, validating the consistency of our approach.

We also plotted a **scatter plot** of the two indices, as attached below.

3 Results

3.1 Judiciary Health Index Across Africa (2023)

The computed **Judiciary Health Index** was visualized using a **choropleth map** (Figure 1), displaying judicial health variations across African countries. The interactive version can be accessed at:

View the (GitHub) hosted map here

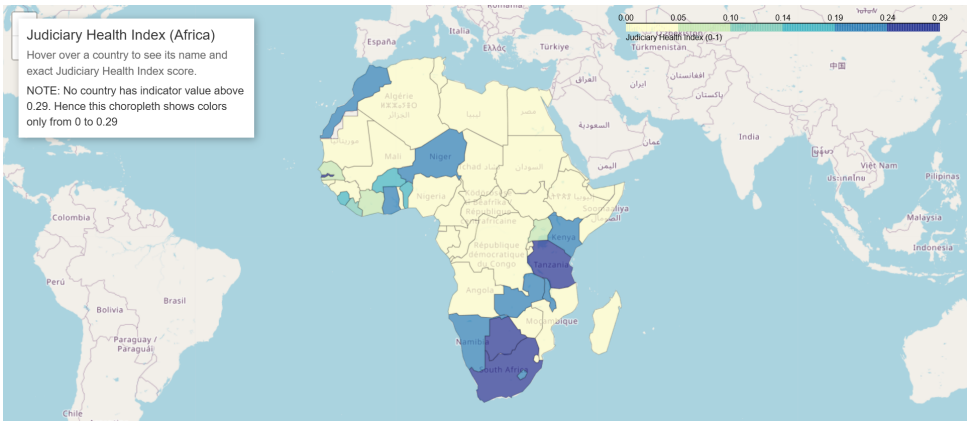

Figure 1: Screenshot of the Judiciary Health Index Choropleth Map

3.2 Judiciary Trends in Selected Countries (2001–2023)

A time-series analysis was conducted for India, Tanzania, Zanzibar, Japan, and Germany to track judiciary health variations over time (Figure 2).

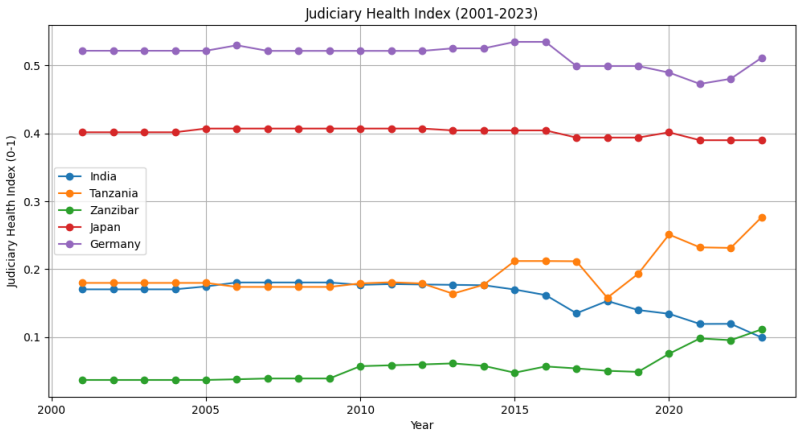

Figure 2: Judiciary Health Index (2001-2023) for Selected Countries

3.3 PCA-Based Index vs. Weighted Index

A PCA-based Judiciary Health Index was computed and compared against the weighted index. The scatter plot (Figure 3) illustrates the relationship between both indices, showing a Pearson correlation coefficient of **0.8837**.

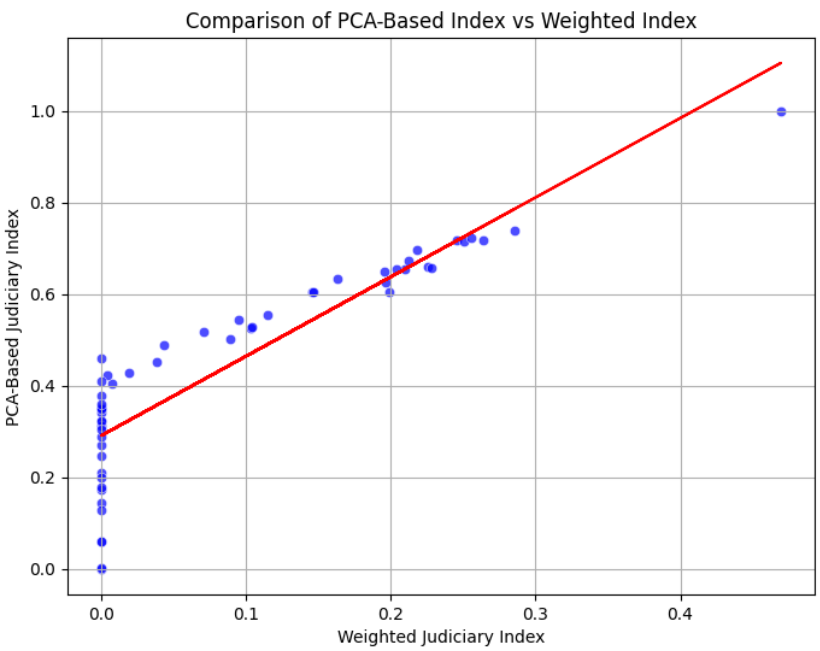

Figure 3: Comparison of PCA-Based Judiciary Health Index vs. Weighted Index

4 Observations and Deductions

4.1 Judiciary Health Across Africa (2023)

Looking at the judiciary health index across Africa, we see clear differences between countries. Some nations have relatively independent courts, while others show clear signs of judicial weakness due to political interference or corruption. The map in the results section highlights these variations, with scores ranging from 0 to 0.29, as no country exceeded that upper limit.

Certain countries, like **South Africa, Namibia, and Botswana in the south and Kenya and Tanzania in the East**, stand out with stronger judiciary health, mostly due to their more stable democratic institutions.

On the other hand, **Central and West African nations** tend to score lower, often reflecting weaker legal systems and external pressures on judicial bodies. This pattern suggests a direct link between how democratic a country is and judiciary strength -

Where democracy thrives, Judicial independence tends to be stronger

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4.2 Judiciary Trends Over Time (2001-2023)

The time-series graph in the results section reveals that countries like **Germany and Japan** have had consistently strong judiciary systems, showing very little fluctuation. This aligns with their long-standing legal traditions and stability.

However in India and Tanzania there are fluctuations from around 2014. In India, the lowering of the index corresponds with election of BJP.
On the other hand, in Tanzania, April 2014 marks the time when the citizens of Tanzania successfully sought changes in the constitution, dividing the government into three parts; Tanzania Mainland, Tanzania Island, and the Republic of Tanzania. [1] They also reduced and reinstated the power of the president.
In Zanzibar, the Judicial strength has increased in 2010 when Zanzibar's First National Unity Government Formed in October 2010 [2].
We see that it has a slight drop in 2015, resulting from it's country, Tanzanian government's division into 3.

4.3 Comparison of PCA and Weighted Indices

To check whether our weighted index was effective, we compared it with a PCA-based judiciary health index. The scatter plot in the results section shows a strong correlation between the two, with a Pearson coefficient of **0.8837**. This suggests that both methods capture similar patterns in judiciary health, reinforcing the reliability of the weighted approach.

However, there are a few cases where PCA assigns slightly higher scores to some countries. This happens because PCA captures additional variance that isn't explicitly factored into our weighted model. Essentially, PCA picks up on patterns that may not be immediately obvious but still influence judicial performance.

4.4 Final Deductions

From all of this, a few things become clear to us:

- Countries with stronger democratic institutions tend to have healthier judiciaries.
- Judiciary health is not static—it changes over time due to legal reforms, political shifts, or internal divisions (like in Tanzania).
- The PCA and weighted indices align well, that is our weighted index is a solid measure of judiciary health.
- Some countries show unexpected rankings in PCA, showing that there may be hidden factors influencing judiciary performance.

— Thank you —

[1] from participedia.net
[2] from crisisgroup.org